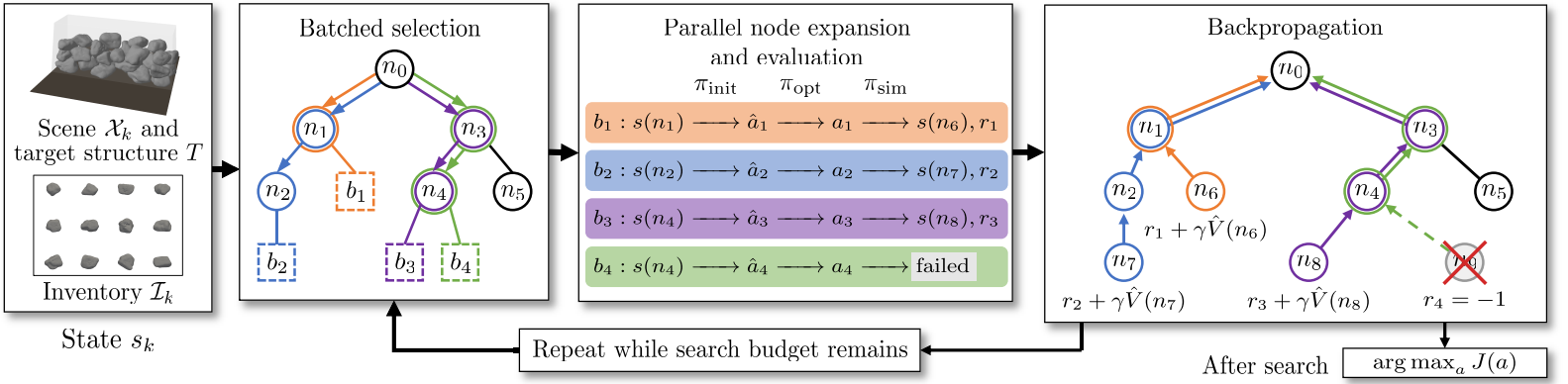

Scene $\mathcal{X}_k$ and target structure T
Inventory $\mathcal{I}_k$
State s_k
Batched selection
n_0
n_1
n_2
b_1
n_3
n_4
n_5
b_2
b_3
b_4
Parallel node expansion and evaluation
π_{init} π_{opt} π_{sim}
$b_1 : s(n_1) \longrightarrow \hat{a}_1 \longrightarrow a_1 \longrightarrow s(n_6), r_1$
$b_2 : s(n_2) \longrightarrow \hat{a}_2 \longrightarrow a_2 \longrightarrow s(n_7), r_2$
$b_3 : s(n_4) \longrightarrow \hat{a}_3 \longrightarrow a_3 \longrightarrow s(n_8), r_3$
$b_4 : s(n_4) \longrightarrow \hat{a}_4 \longrightarrow a_4 \longrightarrow$ failed
Backpropagation
n_0
n_1
n_3
n_2
n_6
n_4
n_5
$r_1 + \gamma \hat{V}(n_6)$
n_7
n_8
n_9
$r_2 + \gamma \hat{V}(n_7)$
$r_3 + \gamma \hat{V}(n_8)$
$r_4 = -1$
Repeat while search budget remains
After search
$\arg\max_a J(a)$